

VoiceCRM AI Agent

Browser Voice Lead Qualification + Make.com CRM Automation

Project documentation for GitHub portfolio and Solutions Engineer - Voice AI applications

1. Summary

VoiceCRM AI Agent is a browser-based voice intake and CRM automation MVP. It captures a customer requirement through microphone input, converts speech to text in the browser, sends the transcript to a FastAPI backend, qualifies the lead, stores the record, and triggers a Make.com webhook to log the lead into Google Sheets.

The project demonstrates practical implementation skills needed for a Voice AI Solutions Engineer role: API design, webhook routing, CRM-ready payloads, automation setup, voice demo design, dashboarding, and fallback handling.

The current demo uses browser voice instead of real phone calls. The backend is structured so it can later connect to Twilio Voice, HubSpot, Zoho, WhatsApp Business API, or live voice-agent APIs.

2. System Flow

Browser voice input -> Speech-to-text transcript -> FastAPI backend -> Lead qualification -> SQLite storage -> Make.com webhook -> Google Sheets -> Streamlit dashboard

3. What We Built

- FastAPI backend with health check, API docs, and browser voice endpoint.
- Browser voice demo page using speech-to-text and text-to-speech.
- Lead qualification response with score, priority, next action, and agent reply.
- Make.com webhook integration to send qualified lead data to an automation scenario.
- Google Sheets output for CRM-ready lead logging.
- SQLite database for local lead/call storage.
- Streamlit dashboard for reviewing captured leads.
- Optional Hugging Face AI layer with rule-based fallback if API access fails.
- Documentation, setup steps, demo script, and GitHub instructions.

4. Implementation Status

Component	Status	Notes
Browser voice input	Working	Uses browser microphone and speech-to-text.
FastAPI backend	Working	Serves APIs and /voice-demo page.
Lead qualification	Working	Rule-based by default; optional Hugging Face mode supported.

Make webhook	Working	Backend sends payload to Make.com webhook.
Google Sheets logging	Working	Rows are created in VoiceCRM Leads sheet.
SQLite storage	Working	Stores local call/lead records.
Streamlit dashboard	Working	Displays stored lead data.
Real CRM writeback	Future scope	HubSpot/Zoho can be added later.
Real phone calls	Future scope	Twilio Voice integration can be added later.

5. Screenshot Evidence

Browser voice demo

VoiceCRM AI Agent — Browser Voice Demo

This is the free voice layer: speak into your laptop microphone, convert speech to text in the browser, send the transcript to FastAPI, qualify the lead, trigger Make, and hear the agent reply using browser text-to-speech.

Stopped listening. You can edit the transcript before sending.

Caller Details

Customer name

Phone

Aisha Khan

+971501112233

Email

Company

aisha@example.com

Palm Clinic

Industry

Preferred follow-up

AI Result

Lead ID: 5

Score: 90

Priority: High

Make: sent

Agent reply

Thanks Aisha Khan. I captured your request. This is a high priority lead with a score of 90. Next step: Book technical discovery call and prepare vertical-specific POC.

Make.com connection/scenario

Google Sheets

Connection *

Creating a connection...

For more information on how to create a connection to Google Sheets, see the [online Help](#).

Cancel

Save

Webhooks

1

Custom webhook

Create a connection

Connection name *

My Google connection

Make's use and transfer of information received from Google APIs to any other app will adhere to [Google API Services User Data Policy](#).

Advanced settings

Close

Sign in with Google

Google Sheets

2

Add a Row

Google Sheets output

VoiceCRM Leads											
File Edit View Insert Format Data Tools Extensions Help											
100% 123 Default... 10 B I A											
B6	fx										
	A	B	C	D	E	F	G	H	I	J	K
1	lead_id	call_id	created_at	customer_name	phone	email	company	industry	requirement	budget	timeline
2	2	2	1780681802	Aisha Khan	971501112233	aisha@example.	Palm Clinic	Healthcare	We need an AI v	12000 AED mon	Go live in 3 we
3	5	5	1780687842	Aisha Khan	971501112233	aisha@example.	Palm Clinic	Healthcare	We are a real es	12000 AED mon	Go live in 3 we
4											
5											
6											
7											

100% 123 Default... 10 B I A											
fx											
K	L	M	N	O	P	Q	R	S	T		
timeline	preferred_chann	lead_score	priority	next_action	workflow	integrations_nee	estimated_comp	recommended_p	crm_status		
Go live in 3 weel	WhatsApp follow	100	High	Book technical d	Appointment boc	Voice telephony	Medium	Build a 5-minute	created		
Go live in 3 weel	WhatsApp follow	90	High	Book technical d	General voice as	Voice telephony	Medium	Build a 5-minute	created		

6. Setup Summary

1. Create a virtual environment and install requirements.txt.
2. Copy .env.example to .env and add MAKE_WEBHOOK_URL.
3. Start backend: `uvicorn backend.app.main:app --host 0.0.0.0 --port 8000 --reload`.
4. Open /voice-demo and test with the sample voice script.
5. Open Make.com scenario and confirm Webhook -> Google Sheets runs successfully.
6. Run Streamlit dashboard: `streamlit run dashboard/app.py`.
7. Take screenshots and save them under docs/images/ before final GitHub push.

7. Environment Variables

```
APP_NAME=VoiceCRM AI Agent
DATABASE_URL=sqlite:///./voicecrm.db
MAKE_WEBHOOK_URL=your_make_webhook_url_here
MAKE_WEBHOOK_SECRET=demo-secret-change-me
CRM_PROVIDER=mock
HUBSPOT_ACCESS_TOKEN=
MOCK_MODE=true
AI_PROVIDER=rules
HF_TOKEN=
HF_MODEL=openai/gpt-oss-120b:fastest
HF_BASE_URL=https://router.huggingface.co/v1
```

8. What Is Real vs Mock

Real: browser voice input, speech-to-text, text-to-speech, FastAPI backend, Make webhook, Google Sheets row creation, SQLite storage, and Streamlit dashboard.

Mock/future scope: direct HubSpot/Zoho writeback and real Twilio phone calls. The current phone-call experience is simulated through the browser voice page.

9. GitHub Push Commands

```
git status
git rm --cached .env # only if .env was accidentally staged
git add .
git commit -m "Add VoiceCRM AI Agent documentation and screenshots"
git push
```

10. Resume Bullet

Built VoiceCRM AI Agent, a browser-based AI voice lead qualification and CRM automation system using FastAPI, speech-to-text, SQLite, Streamlit, Make.com webhooks, and Google Sheets to capture spoken customer requirements and route qualified leads into CRM-ready workflows.

11. Interview Explanation

I built this project to simulate the implementation lifecycle of a Voice AI Solutions Engineer. The browser captures a customer requirement through voice input, converts it to text, sends it to a FastAPI backend, qualifies the lead, stores the record, and triggers a Make.com webhook. Make then logs the qualified lead into Google Sheets. The backend is structured so it can later connect to Twilio for real phone calls and HubSpot or Zoho for CRM updates.